

Austin Senna Wijaya

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EDUCATION

Columbia University

Bachelor of Science: Computer Science

New York, NY

Aug 2024 - May 2028

- GPA: 4.22/4.0; Full-Tuition Scholarship; Dean's List
- Relevant Coursework: Systems Programming, Data Structures & Algorithms, Natural Language Processing, Linear Algebra & Probability, Competitive Programming.

EXPERIENCE

Software Engineer Intern

Jun 2026 – Present

Garner Health

New York, NY

- Engineered a full-stack observability pipeline with automated dashboard and alert provisioning, implementing distributed tracing across Datadog RUM/Replay and OpenTelemetry (Loki, Prometheus, Tempo) to correlate frontend sessions with backend traces.
- Architected an autonomous on-call triage AI agent that ingests cross-stack telemetry to execute root-cause analysis and autonomously generate verifiable fix pull requests adhering to zero-PHI guardrails, resolving 10+ production incidents.

Data Agents Research Assistant

Dec 2025 – Present

Columbia Data, Agents, and Processes Lab

New York, NY

- Engineered a high-throughput benchmark evaluation harness to process a **9.5 TB / 40M-document** S3 data lake, utilizing parallel sandboxed workers and isolated **DuckDB/Python** runtimes, as well as context engineering for multi-agent pipelines.
- Built scalable BM25 and hybrid retrieval APIs utilizing **LanceDB** vector search, full-text indexes, and **NetworkX** graph traversal to support massive-scale source discovery.
- Architected a multi-axis telemetry and ablation framework to capture tool-call traces and agent reasoning and route API failures.

Machine Learning Research Assistant

Dec 2025 – Jun 2026

Columbia Zuckerman Institute

New York, NY

- Compressed **Qwen2-Audio 4096-D** hidden states into **10-100D** feature-specific representations, each preserving one linguistic or paralinguistic feature such as emotion or reasoning while suppressing other features.
- Built and parallelized a GPU-optimized **PyTorch** pipeline for Qwen2-Audio inference and hidden-state extraction, bottleneck-width sweeps, and supervised classification under cross-validation across **12** speech attributes and **120K+** audios.

Data Engineering Intern

Jun 2025 – Aug 2025

Ruangguru (Indonesia's largest edtech platform)

Jakarta, Indonesia

- Engineered scalable ETL pipelines to aggregate millions of asynchronous payment events into structured session data using **BigQuery**, supporting high-throughput downstream analytics.
- Architected an automated data ingestion and sentiment analysis service utilizing **Puppeteer** and **Hugging Face** APIs, integrating the pipeline into live operational monitoring dashboards.

PUBLICATIONS

- [1] *LakeQA: A Benchmark for Complex Exploratory QA over a Million-Scale Data Lake*, **ICML 2026**: Fourth author.
- [2] *SANA: What Matters for QA Agents over Massive Data Lakes?* **VLDB DASHSys 2026**: First author.
- [3] *MINA: Linear Probes Reveal Coding-Sequence Family Signal in Frozen DNA Encoders*, **iscbAI'26**: First author. (Under-review).
- [4] *Targeted Embeddings: Neural Speech Encoding with Large Audio-Language Models*, **NeurIPS 2026**: Third author (Under-review).
- [5] *Be CARE-ful with Text-to-SQL Benchmarks*, **NeurIPS 2026**: Sixth author. (Under-review).

PROJECTS

ResearcherX: Local-First AI Research IDE

[Website](#)

- Built ResearcherX, a local-first AI research editor with **Electron/Next.js/FastAPI** stack and graph-based RAG in **LanceDB** to produce cited answers and detect unsupported claims, contradictions, or citation drift in drafts with model-agnostic AI Agents.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++, Java, JavaScript/TypeScript
- **Web & App:** Next.js, React, FastAPI, Electron, Litestar, OpenTelemetry, Tailwind CSS
- **AI & Data:** Strands Agents, Langchain, Pandas, NumPy, PyTorch, Puppeteer/Selenium, BeautifulSoup
- **Cloud & Databases:** AWS (EC2, S3, Bedrock), Google Cloud (BigQuery, Looker Studio), PostgreSQL, Grafana
- **Tools:** GitHub, Claude Code, Codex, Docker, Copilot, Arduino, Notion

HONORS

- **Databricks Grounded Reasoning Cup 2026:** Columbia representative; worked with Google.
- **Jane Street Mystery Planet Trading & Puzzle Competition 2026:** 1st Place.
- **Hack@Brown 2026:** Finalist; Strongest Product Thinking award; worked with Google Ventures and Partiful.
- **Tencent Kepler Plan S3 Competition 2025:** 2nd Place; Best Code; Best Product Idea; Popularity Award.
- **Indonesia Maju Scholarship:** Full-ride scholarship recipient, 1 of 350 recipients.
- **55th International Chemistry Olympiad 2023:** Final stage, 10th rank for Indonesian national team selection.